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System Integration and Architecture

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EXECUTIVE SUMMARY

1.1 Project Overview

NASync is a comprehensive tracking system designed for the Office of Admissions and Scholarships (OAS) to automate the monitoring of student scholar (NAS) duty hours. It replaces manual paperwork and messaging with a structured digital workflow. Scholars record their time-in and time-out via a mobile application, and department heads verify and approve entries in real time.

The primary goal of NASync is to reduce administrative workload, eliminate human errors in duty-hour computation, and provide transparent, real-time visibility of attendance records for all users. The system is built using Java Spring Boot for the backend API, React (with Vite) for the web application, and Kotlin for the Android mobile application (developed in Android Studio Iguana). This technology stack ensures a reliable, responsive, and maintainable cross-platform solution.

1.2 Objectives

1. Develop an automated time-tracking system that replaces manual logs with accurate time capture and built-in validation mechanisms.

2. Implement a role-based workflow supporting Admin, Department Head, and NAS roles with distinct permissions for scheduling, monitoring, and approvals.

3. Enable real-time system updates to provide live notifications (e.g., approval status changes) across connected clients.

4. Integrate secure authentication mechanisms using JWT for stateless and scalable security, along with Google OAuth for streamlined user authentication

5. Ensure full compliance with IT342 final implementation requirements, including authentication, CRUD operations, system integrations, and other specifications outlined in the project guidelines

1.3 Scope

Included Features:

- User registration and authentication using JWT and BCrypt password hashing
- Secure login functionality with token-based authentication
- Duty scheduling management for assigned personnel
- Time-entry CRUD operations (create, view, update, and delete time logs)

- Role-based approval workflows for submitted time entries
- Email notifications for account registration (welcome email) and approval status updates
- Basic data reporting, including summaries of hours late and absences

Excluded Features:

- Integrate with biometric devices
- Payment processing functionality (not applicable to the system)
- Advanced analytics and data visualization beyond basic reporting summaries
- Complex performance dashboards or predictive reporting features
- Third-party payroll system integration
- Mobile push notifications
- Extended system customization beyond the defined MVP requirements

1.0 INTRODUCTION

1.1 Purpose

This document serves as the comprehensive design specification for the NASync system. It provides detailed requirements, architectural decisions, API contracts, database design, and implementation roadmap to guide development and ensure all components integrate smoothly.

2.0 FUNCTIONAL REQUIREMENTS SPECIFICATION

2.1 Project Overview

Project Name: NASync
Domain: Academic Scholarships Duty Management
Primary Users: OAS Admins, NAS (Scholars), Department Heads

Problem Statement: Manual processing of biometric data, individual messaging, and makeup hour calculations create administrative burden, errors, and lack of transparency.
Solution: Digital time logging system with real-time approvals and tracking dashboards.

2.2 Core User Journeys

Journey 1: OAS Admin - Complete Scholar Onboarding

1. Admin logs into web dashboard
2. Navigates to "Register Scholar" and enters personal info (name, student ID, contact)
3. Configures duty schedule: Morning (8-12), Afternoon (1-5), late threshold (10min), absent (1hr)
4. Assigns department and department head
5. System generates default password and sends welcome email
6. New scholar appears in active tracking list

Journey 2: NAS - Duty Logging (Mobile)

1. NAS downloads app and receives login credentials via email
2. Logs in
3. Sees personalized duty schedule
4. During duty, selects “Regular Duty”
5. Click Time-in button
6. System validates against schedule
7. Shows “Present” if on-time or “Late” if >10 mins passed the scheduled duty

Journey 3: Department Head - Daily Duty Review & Approval (Web)

1. Head logs into web dashboard
2. Sees "Pending Approvals: 5 duties"
3. Filters by NAS or date
4. Views time-in details (scheduled vs actual)
5. For each duty: [APPROVE] [REJECT] [VIEW NOTES] buttons
6. Selects APPROVE
7. Adds optional comments
8. NAS receives notification
9. Duty moves to "Approved History"

10. Late/absent metrics update automatically

Journey 4: NAS - Makeup Duty Process (After Being Late)

1. NAS views personal dashboard
2. Sees "Late: 2.5hrs on Mon and Need 2.5hrs makeup"
3. Selects "MAKEUP DUTY"
4. System shows available makeup slots based on schedule
5. Clocks in for makeup
6. After completion
7. Status updates to "MAKEUP CLEARED"

Journey 5: Overtime Request & Approval (Special Case)

1. NAS needs extra hours
2. Selects "OVERTIME" duty type
3. Clocks in for overtime
4. Department head receives "Overtime" time-in
5. Head reviews
6. APPROVES with notes
7. NAS sees approved overtime
8. Counts toward makeup hour requirements

2.3 Feature List (MoSCoW)

MUST HAVE

1. Complete user authentication with role-based dashboards
2. Scholar registration with full duty schedule configuration
3. Time-in/out with 3 duty types and schedule validation
4. Admin account management
5. Late/absent/make-up hour calculations

SHOULD HAVE

1. Email notifications (welcome, approvals, makeup reminders)
2. Real-time approval notifications
3. Department-level performance summaries

COULD HAVE

1. Bulk approval for heads
2. Duty history with filtering/export
3. Overtime request workflow
4. Makeup duty slot suggestions

WON'T HAVE

1. Biometric integration
2. Complex scheduling conflicts
3. Payroll export

2.4 Detailed Feature Specifications

Feature: User Authentication

- **Screens:** Registration(Admin only), Login
- **Fields:** Email, Password, Confirm Password, Role, ID Number
- **Validation:** Email format, password strength, uniqueness
- **API Endpoints:** POST /auth/register, POST /auth/login, POST /auth/logout
- **Security:** JWT tokens, password hashing with BCrypt

Feature: Scholar Registration (Admin)

- **Screens:** Registration
- **Fields:** Personal info, duty hours (morning/afternoon), late threshold (10min), absent threshold (1hr), schedule, department, head assignment
- **API Endpoints:** POST /scholars, GET /scholars, PUT /scholars/{id}

Feature: Time Logging (NAS)

- **Screens:** Clock in/out
- **Functions:** Clock in/out, duty type selection (regular/makeup/overtime)
- **Validation:** Schedule-based late/absent detection
- **API Endpoints:** POST /duties/clock-in, POST /duties/clock-out

Feature: Approval Workflow (Head)

- **Functions:** View assigned NAS duties, approve/reject, view history, revert decisions
- **API Endpoints:** GET /duties/assigned, PUT /duties/{id}/approve